

MAINNET PREPARATION REPORT

UNIFYVAULT V2 — MAINNET PREPARATION & AUDIT REPORT

Target Network: Base Mainnet (Chain ID: 8453)

Current Active Environment: Base Sepolia (Chain ID: 84532)

Repository: `/var/www/UnifyVault-UV`

Git Commit: `0c85935613ff2c84a7b7e26792187cbd68b3b4f6`

1. Repository Integrity

- Solidity Source Baseline**: `packages/protocol/src/` is completely pristine with **0** modifications and **0** untracked files.
- Test Integrity**: Zero tests were modified or weakened. All 715 Foundry tests and 974 Vitest frontend tests are active and passing.
- Secrets Management**: Zero private keys or mnemonics are tracked in Git. Only `.env.example` templates are checked into version control.

2. Current Single-EOA Status

- Status**: **PASS — FULLY OPERATIONAL**
- Operational Reality**: UnifyVault V2 operates under Single-EOA governance mode today.
- Authorized Authority**:
 - `Master Admin EOA` (`0xe37b77ca9e49c2586365e7394f0f037901ed8a95`) holds `DEFAULT_ADMIN_ROLE` across core protocol contracts.
 - `Deployer & Guardian EOA` (`0x441dbf8076d0b143EC17199baE94Daa884161454`) holds `GOVERNANCE_ROLE` & `GUARDIAN_ROLE` across ProtocolDirectory, UVBE, StakingVault, and controllers.
 - `Timelock Controller` (`0x610c5f66d99993d444561d270fba172db1f7cff1`) enforces 48-hour execution delay.
- Frontend UX**: Dynamic role inspection via `hasRole()` allows single-wallet administrators to manage the protocol without a multisig dependency.

3. Optional Multisig Status

- Status**: **PASS — ZERO REDEPLOYMENT TRANSITION CAPABLE**
- Architecture**: All protocol modules inherit OpenZeppelin `AccessControl` and `Ownable`.
- Quorum Flexibility**: Supports any arbitrary `$M$-of-$N$` configuration (`$1$-of-$1$`, `$2$-of-$3$`, `$3$-of-$5$`, `$4$-of-$7$`).
- No Hardcoding**: No Safe contract address, signer count, or threshold is hardcoded in smart contracts or frontend logic.

4. Contracts Requiring No Code Changes

All 22 contracts in `packages/protocol/src/` require **0** source modifications for Base Mainnet:

- `ProtocolDirectory.sol`

2. `UnifyVaultControllerUpgradeable.sol`
 3. `CustodyVault.sol`
 4. `Treasury.sol`
 5. `OracleManager.sol`
 6. `ChainlinkOracleProvider.sol`
 7. `StrategyManager.sol`
 8. `LiquidityManager.sol`
 9. `CostBasisManagerV2.sol`
 10. `PerformanceManager.sol`
 11. `FeeManager.sol`
 12. `PortfolioManager.sol`
 13. `SwapAdapter.sol`
 14. `UVBTCETHToken.sol`
 15. `P2PEscrow.sol`
 16. `Marketplace.sol`
 17. `P2PReputation.sol`
 18. `GasTreasury.sol`
 19. `UnifyVaultPaymaster.sol`
 20. `UnifyVaultTimelock.sol`
 21. `UVBESstakingVault.sol`
 22. `UVBERewardDistributor.sol`
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5. Configuration Requiring Mainnet Values

- **Parameter:** Base Sepolia Value | Target Base Mainnet Value | Configuration Method
 - ****Chain ID**:** `84532` | `8453` | `NEXT\_PUBLIC\_ACTIVE\_CHAIN="base"`
 - ****Native USDC**:** `0x036CbD53842c5426634e7929541eC2318f3dCF7e` |
`0x833589fCD6eDb6E08f4c7C32D4f71b54bdA02913` | Verified in `constants/index.ts`
 - ****cbBTC**:** `0xB0B47F113Bcab2b0e49fD5d3Bd2CC0e9Aa408b29` |
`0xcbB7C0000aB88B473b1f5aFd9ef808440eed33Bf` | Verified in `constants/index.ts`
 - ****WETH**:** `0xd116ab1c943cf15904eC4c8dd701086f175FA323` |
`0x4200000000000000000000000000000000000000` | Verified in `constants/index.ts`
 - ****USDC/USD Feed**:** Testnet Mock Feed | `0x7e860098F58bBFC8648a4311b374B1D669a2bc6B` | Configured in `DeployMainnet.s.sol`
 - ****cbBTC/USD Feed**:** Testnet Mock Feed | `0x8C74B2811D2F1aD65517ADB5C65773c1E520ed2f` | Configured in `DeployMainnet.s.sol`
 - ****ETH/USD Feed**:** Testnet Mock Feed | `0xe6eb5B9b85cFF2C84Df3De6e7855bC9E76f034d5` | Configured in `DeployMainnet.s.sol`
 - ****Uniswap V3 Router**:** Testnet Mock Router | `0x2626664c2603336E57B271c5C0b26F421741e481` | Configured in `DeployMainnet.s.sol`
 - ****RPC Endpoint**:** `https://sepolia.base.org` | Dedicated Alchemy/Infura URL | `.env.production.local`
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6. Relay & Infrastructure Requirements

- Dedicated ERC-4337 Bundler RPC endpoint (Pimlico / Alchemy / Biconomy) configured for Base Mainnet.
 - Automated gas fee escalation policy for UserOperations in mempool.
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7. Cloud KMS / HSM Signer Requirements

- ****Zero Plaintext Keys**:** Production Paymaster verification signing must be backed by AWS KMS `ECC\_SECG\_P256K1` or Google Cloud HSM.
 - Strict IAM role policies restricting signing capabilities to authorized backend container instances.
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8. Paymaster & Policy Requirements

- Target contract allowlist restricted strictly to Mainnet `USDC`, `UnifyVaultController`, `UVBEToken`, and `P2PEscrow`.
 - Zero native ETH value transfers permitted (`value == 0`).
 - UserOp execution gas cap and fee per gas caps enforced on-chain.
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9. Gas Treasury Reserve Requirements

- Standalone native ETH balance for Paymaster deposit replenishment.
 - Rate limits: `maxRefillPerTx = 0.5 ETH`, `dailyRefillLimit = 2.0 ETH`.
 - Prometheus/Webhook alerting configured when reserve $\leq 1.0 \text{ ETH}$.
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10. Timelock Governance Status

- Delay set to **172,800 seconds (48 hours)**.
 - Operation ID hashing matches OpenZeppelin formula.
 - Frontend operation salt generator hardened with `globalThis.crypto.getRandomValues()`.
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11. Security Findings

- **P0 Critical: 0**
 - **P1 High: 0**
 - **P2 Medium: 0**
 - **P3 Low / Operational Pre-Flight: 2**
 - Provision Cloud KMS signer for production relayer backend.
 - Deploy contracts to Base Mainnet and record live addresses.
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12. Regression Test Results

- **TypeScript Typecheck: 0 Errors** (`tsc --noEmit`).
 - **Web Frontend Vitest Tests: 974/974 Passed** across 80 test files.
 - **Next.js Production Build: 46/46 Pages Compiled** successfully.
 - **Foundry Protocol Suite: 715/715 Passed** across 86 test suites.
 - **On-Chain Negative Simulation Tests: 10/10 Reverted as expected** on Base Sepolia.
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13. Mainnet Launch Blockers (Pre-Flight Operational Tasks)

1. **[OPERATIONAL]:** Cloud KMS signer provisioning for production AA paymaster.
 2. **[CONFIGURATION]:** Dedicated Mainnet RPC & Bundler API keys in production environment.
 3. **[OPERATIONAL]:** Broadcast execution of `DeployMainnet.s.sol` on Base Mainnet (Chain ID: 8453).
 4. **[CONFIGURATION]:** Recording newly deployed mainnet contract addresses into `constants/index.ts`.
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14. Human Governance Decisions Required

1. **Launch Governance Mode:** Choose between launching Base Mainnet initially in **Single Admin EOA** mode (fast bootstrapping, transfer to Safe later) or launching directly into a **Gnosis Safe** multi-sig.
 2. **Launch Liquidity & Caps:** Decide initial deposit caps and rebalance threshold parameters for Day 1 on Base Mainnet.
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15. Exact Next Steps for Production Launch

1. Setup AWS KMS / GCP Cloud HSM key for Paymaster verification signing.

2. Fund deployer wallet on Base Mainnet with deployment gas.
3. Execute ``forge script script/DeployMainnet.s.sol --rpc-url $BASE_MAINNET_RPC --broadcast --verify``.
4. Deploy and fund ``GasTreasury`` and ``UnifyVaultPaymaster`` on Base Mainnet.
5. Populate deployed contract addresses into ``env.production.local`` and ``constants/index.ts``.
6. Deploy frontend to production hosting (Vercel / Cloudflare Pages / AWS).
7. Perform smoke test deposit and verification on Base Mainnet.

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16. Final Production Verdict

\$\$\$ CONDITIONALLY; READY; —; SPECIFIC; PRE - FLIGHT; BLOCKERS; REMAIN}\$\$\$

****Rationale****:

The smart contract codebase and frontend application are ****100% Code-Ready**** with zero defects, zero modified Solidity source files, and 100% passing tests. The conditionality is purely ****operational pre-flight****: deploying the contracts to Base Mainnet, configuring Cloud KMS key management, and setting production RPC credentials.